A Fuzzy Logic Based Vehicle Maintenance Advisory System

Module: CSIP5304 — Fuzzy Logic & Evolutionary Computing
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1 Literature Review

1.1 Fuzzy Logic and How It Applies to Condition Monitoring

Fuzzy logic is commonly used in situations where there is uncertainty or where information is expressed in human terms rather than exact numerical values. For example Vehicle maintenance decisions these are often based on input judgments. In practice a person may describe coolant as “low”, vehicle usage as “high” or brake condition as “old” instead of using numerical values. (Zadeh, 1965; Ross, 2010)

A mechanic doesn't decide that a car suddenly needs a service the moment it crosses exactly 1,000 km. The concern grows gradually as mileage builds up and it's affected by several other things at the same time.

Vehicle maintenance is a good example of where that kind of reasoning matters. A coolant level of 45% is not Dangerously low but it's not fine also it sits somewhere in between and an advisory system should advise by using Standard manufacturer service these are fixed schedules that don't account for how a specific vehicle is actually being used or what condition it's actually in A fuzzy system can look at several degradation koints at once according to rules and arrive at a recommended value. (Jardine et al., 2006; Liao, 2005)

The aim of this project is to develop a fuzzy inference system in MATLAB that advises how urgently a vehicle should be serviced. The system considers three input variables: mileage, coolant level, and brake fluid age. Based on these inputs, it produces a single output called urgency, represented on a scale from 0 to 100. The system is implemented as a Type-1 Mamdani fuzzy inference system.

1.2 The Mamdani inference Model

A Mamdani works step by step in a simple way. First, it fuzzifies the inputs, meaning it checks how much each value belongs to categories like low, medium, or high. Then it applies the rules where each rule is activated depending on how well the inputs match its conditions. (Mamdani and Assilian, 1975)

After that all the rule outputs are combined together. Finally, the system converts this fuzzy result into one clear number called defuzzification which gives the final urgency value which is the process of translating that overlapping mix of fuzzy advice into a single exact number like a final Urgency Score of 71.5 that a driver can actually act on. (Ross, 2010)

The rules themselves are written in plain language 'IF mileage is high AND brakes are old THEN urgency is high' which makes the system easy to explain.

1.3 AND Operators and Defuzzification

When a rule has multiple conditions like "IF mileage is high AND coolant is low" the system has to figure out how strongly to fire that rule. The standard MIN operator handles this by just taking the lowest score out of the conditions. This is a very safe approach because it means a rule is only as strong as its weakest link. For a vehicle

maintenance advisor this is ideal we only want to trigger a severe multi condition warning if all the critical thresholds are actually being crossed. (Klir, G.J. and Yuan, B., 1995)

Another option is the product (prod) operator where the values are multiplied together. This gives a smoother result when both conditions are partly true. However even if one condition is weak, it still contributes instead of limiting the rule completely. Because of this, the product operator can sometimes give slightly stronger outputs than expected. For this kind of system, the minimum operator is usually a better and more cautious choice. (Klir, G.J. and Yuan, B., 1995; Ross, 2010)

After all the rules are applied, the output is still in a fuzzy form. So, the system needs one final step to turn it into a single clear number. This step is called defuzzification.

The most commonly used method is the centroid methad which finds the “centre” of the output. think of it like taking a balanced average of all the possible outputs. This gives a smooth and realistic result which is useful for systems that need gradual decisions.

Another method is Mean of Maximum (MOM). Instead of looking at the whole shape, it only looks at the highest points and takes their average (Hellendoorn and Thomas, 1993)

2 System Design and Implementation

2.1 The Problem

The system is created in such a way that it provides a vehicle owner with a straightforward answer to a simple question: how urgently do I need to book a service? Instead of depending on a set mileage counter or a clock showing a set time, it takes into account three items simultaneously how many miles the vehicle has been driven since the last service, the appearance of the coolant level, and the duration of fitted brake pads.

These three inputs were chosen because they're independent of each other and each captures a different type of degradation. High mileage doesn't tell you anything about coolant level. A car with low mileage might still have three-year-old brake pads but Coolant can drop between services for reasons that have nothing to do with how much the car has been driven. Each input contributes something that the others can't

The output is a continuous urgency score from 0 to 100 mapped to three levels: low (no rush book at your convenience), medium (worth booking in the next few weeks), and high (this needs attention soon)

2.2 Membership Functions

the three inputs is divided into three linguistic terms. The FIS architecture image you provided shows this clearly: mileage, coolant, and brakeAge each use three membership functions, while the output urgency also uses three membership functions.

A combination of trapezoidal (`trapmf`) and triangular (`trimf`) functions were utilized (Ross, 2010; Jantzen, 2013).

Mileage (km)	low	trapmf	[0, 0, 200, 400]	The car hasn't done much since its last service
	medium	trimf	[300, 500, 700]	A normal amount of use getting towards a service
	high	trapmf	[600, 800, 1000, 1000]	Heavy use a service is likely needed
Coolant (%)	low	trapmf	[0, 0, 20, 40]	Danger zone overheating becomes a real risk
	medium	trimf	[30, 50, 70]	Getting low worth keeping an eye on
	normal	trapmf	[60, 80, 100, 100]	Coolant is at a healthy level
Brake Age (months)	new	trapmf	[0, 0, 6, 12]	Brake Pads were recently replaced no concern
	aging	trimf	[10, 18, 26]	Starting to wear an inspection makes sense soon
	old	trapmf	[24, 30, 36, 36]	Brake Pads are past their service life
Urgency (0–100)	low	trimf	[0, 0, 40]	No rush book at next convenience
	medium	trimf	[30, 50, 70]	Worth booking in the coming weeks
	high	trapmf	[60, 80, 100, 100]	Needs attention soon don't leave it

Table 1

2.3 Rule Base

There are eleven rules in total enough to cover the key input combinations without redundancy. Unnecessary rules don't just clutter the design they distort the output Table 2 shows the complete rule base with a plain English explanation of the reasoning behind each one.

#	Mileage	Coolant	Brake Age	Urgency	Why
1	low	normal	new	**low**	Everything looks fine no service needed

2	medium	normal	aging	**medium**	Mileage and brakes both book soon
3	any	low	any	**high**	Safety override critically low coolant
4	high	any	old	**high**	Both main wear indicators are critical act immediately
5	medium	medium	new	**medium**	Mileage is building and coolant is dropping slightly
6	low	medium	old	**medium**	Brakes are old regardless of low mileage get them checked
7	low	normal	old	**medium**	Coolant is fine but those brakes still need attention
8	high	normal	new	**medium**	High mileage alone is enough to warrant a visit
9	medium	medium	aging	**medium**	Nothing alarming but three things are degrading
10	medium	normal	old	**high**	Old brakes with growing mileage don't wait on this one
11	medium	normal	new	**low**	Coolant and brakes are fine

Table 2

Rule 3 is the most important one. It has no conditions on mileage or brake age it fires whenever coolant is critically low. A dangerously low coolant level is a safety issue that shouldn't be softened by anything else. Rule 4 captures the same logic for the two primary wear indicators when both mileage and brake age are critical

2.4 System Architecture

The system is built as a Type-1 Mamdani in MATLAB using the Fuzzy Logic. The baseline uses AND = min, Defuzzification = centroid. Figure 1 shows how MATLAB represents the overall architecture, and Figure 2 shows the complete rule list as it appears in the rule editor. (MathWorks, 2026)

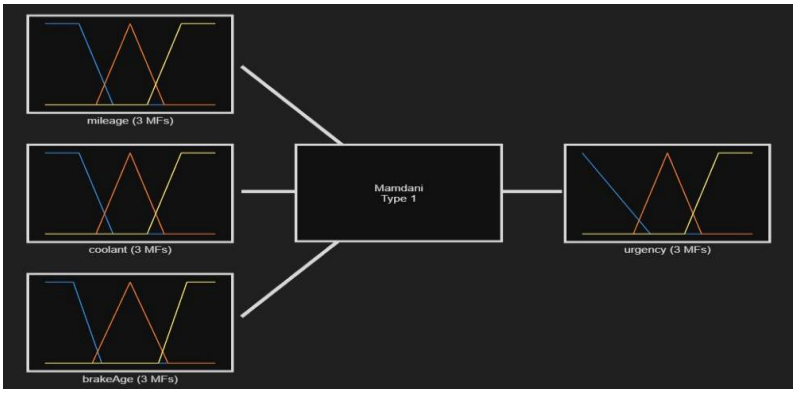


Figure1 three inputs feeding a Type-1 Mamdani FIS with a single urgency output

	Rule	Weight	Name
1	If mileage is low and coolant is normal and brakeAge is new then urgency is low	1	rule1
2	If mileage is medium and coolant is normal and brakeAge is aging then urgency is medium	1	rule2
3	If coolant is low then urgency is high	1	rule3
4	If mileage is high and brakeAge is old then urgency is high	1	rule4
5	If mileage is medium and coolant is medium and brakeAge is new then urgency is medium	1	rule5
6	If mileage is low and coolant is medium and brakeAge is old then urgency is medium	1	rule6
7	If mileage is low and coolant is normal and brakeAge is old then urgency is medium	1	rule7
8	If mileage is high and coolant is normal and brakeAge is new then urgency is medium	1	rule8
9	If mileage is medium and coolant is medium and brakeAge is aging then urgency is medium	1	rule9
10	If mileage is medium and coolant is normal and brakeAge is old then urgency is high	1	rule10
11	If mileage is medium and coolant is normal and brakeAge is new then urgency is low	1	rule11

Figure 2: All eleven rules

3 Testing Across System Variants

3.1 The Three Configurations

To investigate how much the defuzzification method and the AND operator actually matter the same FIS was run in three configurations Table 3 summarises the three setups (Klir and Yuan, 1996)

Table 3: The Three System Configurations

System	AND Operator	Defuzzification	membership functions *2* different in each system
System 1 (Baseline)	Minimum (min)	Centroid	`Trapmf and trimf`
System 2 (MOM)	Minimum (min)	Mean of Maximum (MOM)	`Trapmf and trimf`
System 3 (Product)	Algebraic Product (prod)	Centroid	`Trapmf and trimf`

3.2 Test Cases

Eight test cases were designed to cover different parts of the input space typical cases, safety critical conditions, boundary overlap zones, and moderate midpoint. Each set of inputs was passed to all three systems via MATLAB's evalfis function

- **TC1 [650, 35, 25]:** Most interesting for comparing methods.
- **TC2 [100, 90, 10]:** Everything looks healthy. Expected: low urgency across all systems.
- **TC3 [200, 10, 8]:** Critically low coolant. Rule 3 should dominate regardless of the other inputs.
- **TC4 [150, 90, 5]:** Best case scenario.
- **TC5 [500, 50, 18]**

- **TC6 [800, 80, 32]:** High mileage and old brakes Rule 4
- **TC7 [350, 85, 2]:** Approaching medium mileage but coolant and brakes still fine.
- **TC8 [350, 65, 11]**

3.3 System 1 Results Centroid + Min

Figures 3 and 4 show the MATLAB output for System 1 running TC1 [650, 35, 25]. Figure 3 shows the red marker lines on each input MF, showing exactly where each input value falls relative to the categories. Figure 4 shows the resulting output urgency with the defuzzified centroid result marked plus the 3D surface map showing how urgency varies across mileage and coolant

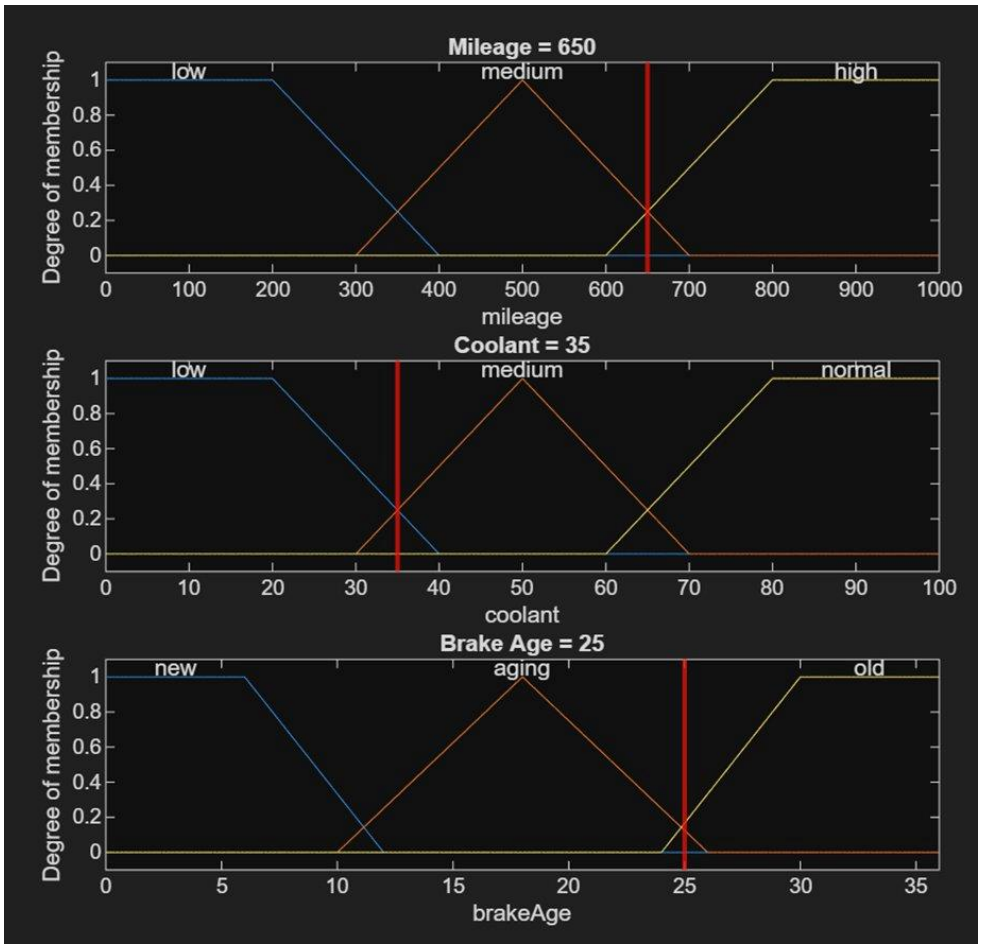


Figure 3 (System 1): TC1 [Mileage=650, Coolant=35, Brakeage=25].

System	AND Operator	Defuzzification	What we're testing
System 1 (Baseline)	Minimum (min)	Centroid	the most widely recommended setup

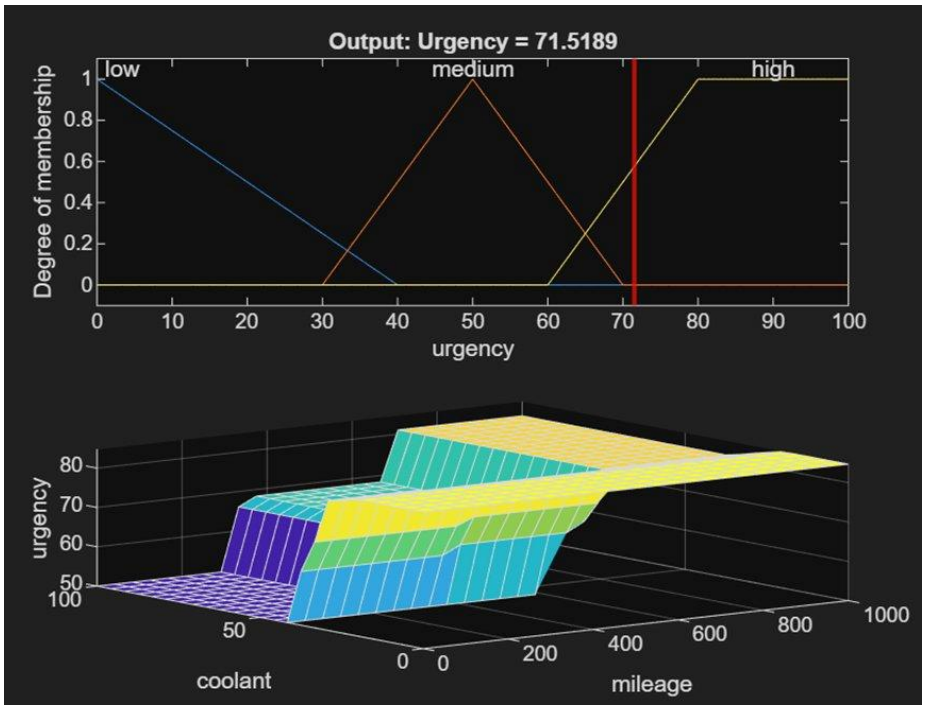


Figure 4 (System 1): Output urgency

the condition is approaching a near critical mainly because the coolant level is low and the brake age is closer to old. These two factors increase the urgency even though the mileage is not at the maximum level.

Among the rules the most influential ones are:

- Rule 3: If coolant is low then urgency is high
- Rule 4: If mileage is high and brake age is old then urgency is high

The centroid calculates the centre of this distribution resulting in an urgency value of 71.52. This indicates a moderately high urgency

3.4 System 2 Results — MOM + Min

Figures 5 show the same TC1 test case run through System 2 where the defuzzification method has changed from centroid to MOM. The input 2 different MFs has been used the input red lines fall in the same places but the output urgency score changes from 71.52 to 82.50

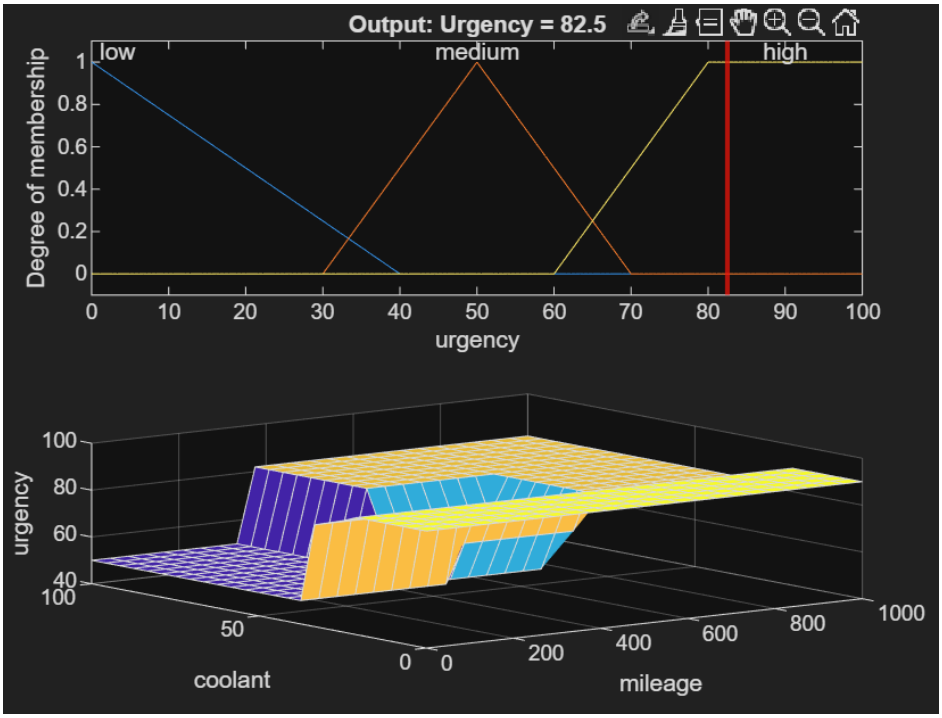


Figure 5 (System 2): Output urgency

The System 2 surface map (Figure 5) versus the smooth gradient in System 1 (Figure 4) is a direct consequence of how MOM works. Because MOM tries to brimg crisp output MF is dominant, transitions between urgency levels produce sharper steps rather than gradual slopes. For the same TC1 inputs, MOM's 82.50 sits clearly within the 'high' value (80–100) while centroid's 71.52 sits at the upper edge of 'medium' a meaningful difference in practice.

3.5 System 3 Results — Centroid + Product

Figures 7 and 8 show System 3, where the AND operator has changed from minimum to algebraic product while centroid defuzzification is retained.

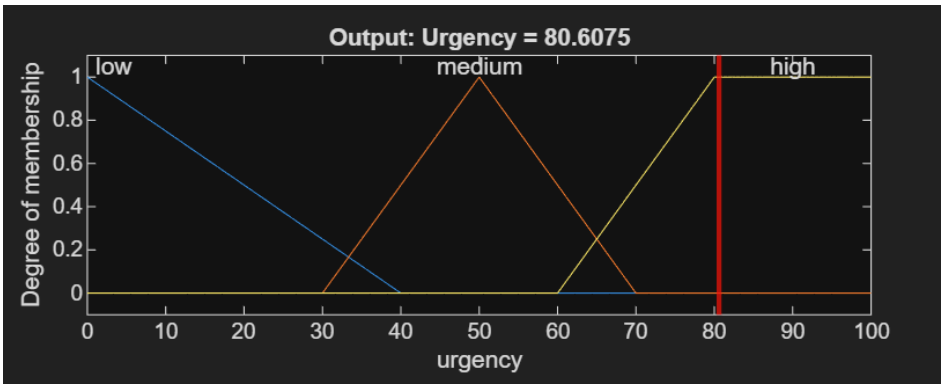


Figure 6 (System 3): Output urgency

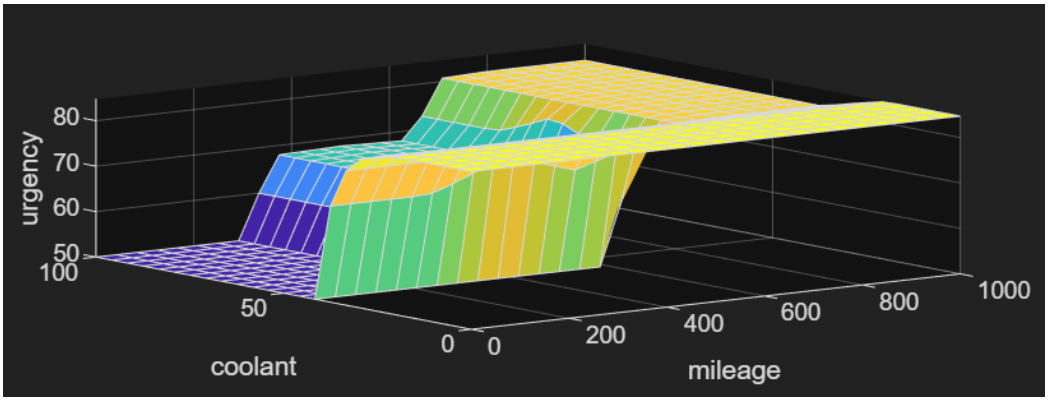


Figure 6 (System 3): Output urgency

Under these conditions System 3 calculated a final Urgency Score of 80.61 The output and 3D control surface map above system retains the smooth graduated transitions characteristic of Centroid defuzzification. There are no harsh, blocky steps like those seen in the MOM method. However, the multiplicative logic of the PROD operator reshapes the slopes and contours of the control surface compared to the baseline system. This confirms that while PROD provides a blended response to overlapping variables, the standard MIN operator remains the more predictable choice

3.6 Full Results Summary

Table 4 shows the urgency scores from all three configurations across all eight test cases.

Table 4: Urgency Scores Across All Configurations and Test Cases

TC	Mileage	Coolant	Brake Age	Sys 1 (Centroid)	Sys 2 (MOM)	Sys 3 (Product)	What's happening
1	650	35	25	**71.52**	**82.50**	**80.61**	Multi-rule overlaps all three inputs
2	100	90	10	**16.63**	**13.00**	**16.63**	All good Rule 1 dominates, low urgency confirmed
3	200	10	8	**84.70**	**90.00**	**84.70**	Critically low coolant fires Rule 3 regardless of anything else
4	150	90	5	**13.00**	**0.00**	**13.00**	Near-perfect condition MOM returns 0.00 (Section 4.1)

5	500	50	18	**50.00**	**50.00**	**50.00**	All inputs at exact MF
6	800	80	32	**84.70**	**90.00**	**84.70**	Rule 4: high mileage plus old brakes
7	350	85	2	**17.37**	**15.00**	**17.37**	Good coolant and new brakes keep urgency firmly low
8	350	65	11	**33.91**	**33.00**	**34.50**	All good

4 Analysis and Discussion

4.1 Centroid vs MOM: Where Do They Differ and Why?

The centroid approach examines the output and identifies point of average value. Thus, rather than considering one powerful rule, it considers all the rules that have been triggered Because of this the output is more smoothly changing and seems more balanced. (Ross, 2005)

Although some of the rules may indicate high urgency and some may indicate medium centroid mixes everything up and provides a reasanable middle value.

The MOM approach on the other hand is different. It just considers the top of the output So which rules are the strongest, determine the output. This causes the output to be more extreme as it does not take into account the smaller contributions of other rules.

This can be clearly seen in the test cases. For example, in Test Case 1, centroid gives around 71.52, while MOM increases it to 82.50. In Test Case 3, MOM pushes the value up to 90.00, while centroid stays slightly lower at 84.70. In Test Case 4, MOM even drops the output to 0.00, while centroid still gives a small but reasonable value of 13.00. These differences show that MOM tends to “jump” to stronger decisions, while centroid stays more stable.

The difference becomes more noticeable when the inputs are in overlap regions where multiple membership functions are active at the same time. In those situations, centroid gives a smooth transition but MOM tends to snap to whichever output is strongest.

In general, in a system such as vehicle maintenance advice the centroid method is more effective since it provides more realistic. MOM can still be useful in comparison, but may also be too severe to be applied in this way. (Hellendoorn and Thomas, 1993; Ross, 2010)

4.2 Minimum vs Product

For six of the eight test cases Systems 1 and 3 produce identical results. Minimum and product only change when at least two inputs in the same rule. If any inputs is at full membership min (a, b) and a × b give the same result. (Klir and Yuan, 1996)

TC1 is where the gap is biggest 71.52 vs 80.61, a difference of just over 9 points. All three inputs are simultaneously in transition zones: mileage activates both medium and high, coolant activates both low and medium, brake age activates both aging and old. Under minimum, each rule's firing strength is capped by its weakest input. And Under product all three input multiply and contribute to the firing strength. The rules pointing toward high urgency particularly Rule 4 (Jantzen, 2013)

TC8 shows a smaller version: 33.91 vs 34.50. The inputs here are only moderately in overlap zones, so this is minor.

For a system with safety, minimum is the more defensible choice.

5 Conclusion

The system behaves as expected across all eight test cases and the three configurations tested. It is able to clearly distinguish between low urgency situations (around 13–17), moderate conditions (around 50), and high-priority cases (above 70). The safety rule for low coolant is especially consistent, always pushing the output toward high urgency when triggered.

When comparing the design choices, the difference in defuzzification methods had the most noticeable impact. The centroid method proved to be the more suitable option as it takes all active rules and produces a smooth, balanced output. and the MOM method tends to jump toward the strongest output leading to more extreme and less gradual results.

The comparison between the AND operators showed much smaller differences. In most test cases, the minimum and product operators produced very similar results. The product operator only resulted in slightly higher urgency values in situations where multiple conditions overlapped. However, for a safety related application, the minimum operator is more appropriat as it limits the rule strength based on the weakest condition and therefore produces a more cautious output.

A natural extension of this system would be to include an additional input, such as engine oil level, to make the model more complete. In the longer term, methods like ANFIS could be used to tune the membership functions automatically using real maintenance data. (Jantzen, 2013; Jardine, Lin and Banjevic, 2006)

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